Highlights

Fill-Dependent Resonance of the Human Urinary Bladder:
Competing Geometry and Stiffening in a Fluid-Filled Viscoelastic Shell

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- Bladder n=2 frequency follows a U-shaped curve: minimum 13.5 Hz at 222 mL
- Minimum arises from competition between geometric softening and detrusor strain-stiffening
- Sub-resonant forced response at 4 Hz to 8 Hz produces larger wall displacements than resonance
- Mechanical (WBV) coupling exceeds airborne acoustic coupling by ~6,400×
- Same analytical framework as abdominal cavity model, with fill-dependent parameters

Fill-Dependent Resonance of the Human Urinary Bladder: Competing Geometry and Stiffening in a Fluid-Filled Viscoelastic Shell

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Abstract

We model the human urinary bladder as a fluid-filled viscoelastic spherical shell whose geometry, wall stiffness, and internal pressure vary with fill volume (50 mL to 500 mL). The fundamental flexural mode ($n = 2$) follows a distinctive U-shaped frequency–volume curve: as the bladder fills, increasing radius and wall thinning lower the frequency (geometric softening), while exponential strain-stiffening of the detrusor muscle raises it. These competing effects produce a minimum of 13.5 Hz at 222 mL—a fill state corresponding to early urge sensation—with higher frequencies at both extremes of the physiological range (13.5 Hz to 17.1 Hz for fills above 150 mL). Because the predicted resonance lies above the ISO 2631 peak pelvic transmissibility band (4 Hz to 8 Hz), the bladder responds to occupational whole-body vibration (WBV) primarily as a forced sub-resonant oscillator. At 6 Hz, where pelvic amplification is maximal, the sub-resonant wall displacement (~160 µm) exceeds the at-resonance displacement (~47 µm) because the larger pelvic transmissibility more than compensates for the reduced modal amplification. Coupling analysis confirms that mechanical vibration transmitted through the pelvis is approximately 6,400 times more effective than airborne sound at exciting bladder wall modes, consistent with the epidemiological dominance of the WBV pathway. These results provide a first-principles physical basis for the fill-dependent resonance behaviour of the bladder wall and identify the sub-resonant forced response as the regime most relevant to occupational WBV exposure.

Keywords: bladder resonance, fill-dependent frequency, fluid-filled shell, whole-body vibration, sub-resonant response, viscoelastic shell

1. Introduction

The natural frequencies of fluid-filled biological shells depend on geometry, wall stiffness, and internal pressure—all of which vary with the organ's functional state. The urinary bladder is a compelling example: as it fills from 50 mL to 500 mL, the radius doubles, the wall thins by a factor of three, the detrusor muscle stiffens five-fold, and the intravesical pressure rises six-fold. These competing effects—some lowering and others raising the natural frequency—produce a rich, non-monotonic frequency–volume relationship that has not previously been characterised from first principles.

Whole-body vibration (WBV) at 4 Hz to 16 Hz is a recognised occupational hazard for vehicle operators. ISO 2631-1 [1] documents peak pelvic transmissibility at 4 Hz to 8 Hz, and epidemiological studies report a consistent association

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between prolonged WBV exposure and lower urinary tract symptoms—particularly increased urgency and frequency—among truck drivers, bus operators, and forklift operators [2, 3]. The association between prolonged driving and urgent bathroom stops is universally recognised, if rarely discussed in the vibration literature with the candour it deserves. Despite this well-established empirical record, no first-principles mechanical model of the urinary bladder exists to explain *why* whole-body vibration should preferentially excite this organ, or to predict how its susceptibility changes with fill state.

The urinary bladder is, from a structural dynamics perspective, a remarkably clean example of a fluid-filled viscoelastic shell. The detrusor muscle forms a roughly spherical wall of thickness 1.5 mm to 5 mm (depending on fill state) enclosing urine—an incompressible fluid with density and bulk modulus essentially identical to water. The geometry, wall stiffness, and internal pressure all vary systematically with fill volume, making the bladder a natural candidate for parametric analysis across its physiological operating range.

In a companion study [4], we developed an oblate spheroidal shell model of the human abdominal cavity to investigate the coupling between airborne infrasound and gastrointestinal resonance. That analysis revealed a full mechanical-to-airborne displacement ratio $\mathcal{R}_{\text{full}} \approx 3.3 \times 10^4$ between mechanical and airborne excitation pathways—a result that explains why whole-body vibration at 4 Hz to 8 Hz causes well-documented gastrointestinal effects while airborne infrasound at comparable frequencies does not. The present work applies the same analytical framework to the bladder, adapting the geometry, material properties, and boundary conditions to this smaller, thinner-walled, and more compliant organ.

The mechanical properties of the bladder wall have been characterised by several groups. Ultrasound bladder vibrometry [5] measured *shear* moduli of 9.6 kPa to 48.7 kPa across fill states from 187 mL to 267 mL, demonstrating pronounced strain-stiffening. For nearly incompressible tissue ($\nu \approx 0.49$), the Young's modulus is $E = 2(1 + \nu)\mu \approx 3\mu$, giving $E \approx 29$ kPa to 145 kPa. Barnes [6] performed tensile testing of excised bladder tissue, reporting Young's moduli from 10 kPa at rest to several hundred kilopascals at high stretch ratios, consistent with the converted vibrometry range. The wall exhibits significant viscoelastic behaviour: stress relaxation, creep, and a loss tangent estimated at 0.3–0.5 from dynamic measurements [7]. Wall thickness decreases from approximately 5 mm when empty to 1.5 mm to 3 mm at capacity [8], consistent with a constant tissue volume assumption. Intravesical pressure follows the characteristic cystometric curve: a compliant phase at low fill (5 cm to 10 cm water column) giving way to steep pressure rise above 300 mL [9].

We apply these data to the fluid-filled shell framework. The bladder is modelled as a spherical shell (the near-spherical approximation is appropriate here, unlike the distinctly oblate abdominal cavity) with fill-dependent radius, wall thickness, elastic modulus, and internal pressure. We compute the flexural mode frequencies across the physiological fill range (50 mL to 500 mL), characterise the non-monotonic frequency–volume relationship, identify the minimum-frequency fill state, quantify the sub-resonant forced response at occupational WBV frequencies, and compare airborne acoustic coupling with mechanical WBV coupling to the bladder wall.

The remainder of this paper is organised as follows. Section 2 presents the mathematical formulation, including the fill-dependent parameter model and the modal analysis. Section 3 reports the fill-dependent frequency behaviour,

elevates the sub-resonant forced response analysis, and quantifies coupling pathways. Section 4 separates model predictions from biological hypotheses and clinical conjecture. Section 5 draws conclusions.

2. Mathematical Formulation

2.1. Model Geometry and Assumptions

The human urinary bladder is idealised as a thin spherical shell of radius R, wall thickness h, completely filled with an inviscid, irrotational fluid of density ρ_f and bulk modulus K_f. Unlike the abdominal cavity—which requires an oblate spheroidal geometry with aspect ratio $c/a \approx 0.67$ [4]—the bladder is approximately spherical at moderate-to-full fill states, making the spherical shell approximation both simpler and more appropriate.

The thin-shell assumption requires $h/R \ll 1$. For the bladder at 300 mL, $h \approx 1.8$ mm and $R \approx 42$ mm, giving $h/R \approx 0.04$, which satisfies the condition comfortably. At low fill volumes, however, the ratio increases: $h/R \approx 0.12$ at 100 mL and $h/R \approx 0.22$ at 50 mL, exceeding the conventional $h/R < 0.1$ threshold. Results below 150 mL should therefore be treated with caution; they are retained in Table 2 for completeness but marked with a dagger. The Love–Kirchhoff hypothesis (normals remain straight and normal to the deformed mid-surface) is applicable for fill states above 150 mL.

The bladder wall is modelled as a Kelvin–Voigt viscoelastic material with complex modulus

$$E^* = E(1 + i\eta), \tag{1}$$

where η is the loss tangent. The quality factor is $Q = 1/\eta$ and the viscous damping ratio is $\zeta = \eta/2$. Published values of η for bladder wall tissue range from 0.3 to 0.5 [7, 6]; we adopt $\eta = 0.4$ ($Q = 2.5$, $\zeta = 0.2$) as a central estimate. The flexural rigidity of the shell is

$$D = \frac{Eh^3}{12(1 - \nu^2)}. \tag{2}$$

2.2. Fill-Dependent Parameters

All geometric and material properties vary with fill volume V (mL). We adopt the following constitutive relationships, drawn from the literature reviewed in §1.

Radius.. The equivalent sphere radius follows directly from the fill volume:

$$R = \left(\frac{3V}{4\pi}\right)^{1/3}, \tag{3}$$

giving $R = 2.3$ cm at 50 mL and $R = 4.9$ cm at 500 mL.

Wall thickness.. We assume approximately constant tissue volume V_{tissue}, calibrated to $h_0 = 5$ mm at $V_0 = 50$ mL. The tissue shell volume is

$$V_{\text{tissue}} = \frac{4}{3}\pi \left[(R_0 + h_0)^3 - R_0^3\right], \tag{4}$$

from which $h(V)$ is obtained by solving

$$(R + h)^3 = R^3 + \frac{3V_{\text{tissue}}}{4\pi}, \tag{5}$$

subject to a physiological floor of $h_{\min} = 1.5\,\text{mm}$. This yields $h = 5.0\,\text{mm}$ at 50 mL, thinning to 1.5 mm at 400 mL and above.

Elastic modulus.. Bladder wall stiffness increases dramatically with stretch (strain-stiffening), a hallmark of collagenous soft tissue [10]. We model this as a log-linear interpolation between vibrometry-derived bounds.

Nenadic et al. [5] report *shear* moduli $\mu = 9.6\,\text{kPa}$ to $48.7\,\text{kPa}$ from ultrasound bladder vibrometry. For nearly incompressible tissue ($\nu = 0.49$), the Young's modulus is

$$E = 2(1+\nu)\,\mu \approx 2.98\,\mu, \tag{6}$$

giving $E \approx 29\,\text{kPa}$ to $145\,\text{kPa}$. These converted bounds are consistent with the tensile-test data of Barnes [6], who reports $E = 10\,\text{kPa}$ to $800\,\text{kPa}$ depending on tissue layer and stretch ratio. We adopt:

$$E(V) = E_{\min}\left(\frac{E_{\max}}{E_{\min}}\right)^{\phi}, \qquad \phi = \frac{V - V_{\min}}{V_{\max} - V_{\min}}, \tag{7}$$

where $E_{\min} = 29\,\text{kPa}$ at $V_{\min} = 50\,\text{mL}$ and $E_{\max} = 145\,\text{kPa}$ at $V_{\max} = 500\,\text{mL}$, with ϕ clipped to $[0, 1]$.

Intravesical pressure.. The pressure–volume relationship follows the characteristic cystometric curve [9]: a compliant phase at low fill followed by a steep rise. We model this as a quadratic:

$$P(V) = P_{\min} + (P_{\max} - P_{\min})\,\phi^2, \tag{8}$$

where $P_{\min} = 490\,\text{Pa}$ ($\approx 5\,\text{cmH}_2\text{O}$) and $P_{\max} = 2940\,\text{Pa}$ ($\approx 30\,\text{cmH}_2\text{O}$).

Table 1: Bladder model parameters at representative fill states.

Parameter	Symbol	50 mL	150 mL	300 mL	500 mL	Unit
Radius	R	2.29	3.30	4.15	4.92	cm
Wall thickness	h	5.00	2.73	1.79	1.50	mm
Elastic modulus	E	29.0	41.5	70.9	145.0	kPa
Intravesical pressure	P	5.0	6.2	12.7	30.0	cmH_2O
Poisson's ratio	ν	0.49 (all fill states)				—
Wall density	ρ_w	1050				kg/m^3
Fluid density	ρ_f	1020				kg/m^3
Fluid bulk modulus	K_f	2.2				GPa
Loss tangent	η	0.40				—

Table 1 summarises the model parameters at four representative fill states. The Poisson's ratio $\nu = 0.49$ reflects the near-incompressibility of soft tissue. The fluid properties correspond to urine at body temperature, which is effectively identical to water for acoustic purposes.

2.3. Breathing Mode ($n = 0$)

The breathing mode is a spatially uniform radial oscillation. As in the abdominal model [4], the restoring force is dominated by the fluid bulk modulus:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\text{shell}} + k_{\text{fluid}}}{\rho_w h + \rho_f R}}, \tag{9}$$

where $k_{\text{shell}} = 2Eh/[R^2(1-\nu)]$ and $k_{\text{fluid}} = 3K_f/R$. For the bladder at 300 mL, $k_{\text{fluid}} \approx 1.6 \times 10^{11}$ Pa/m while $k_{\text{shell}} \approx 2.2 \times 10^4$ Pa/m—a ratio exceeding 10^6. The breathing mode frequency is approximately 9500 Hz, firmly in the ultrasonic range and irrelevant to WBV exposure.

2.4. Flexural Modes ($n \geq 2$)

Flexural modes involve shape changes that preserve the enclosed volume to leading order. The fluid contributes only added mass, not volumetric stiffness. Following Lamb [11] and Junger & Feit [12], the natural frequency of flexural mode n is

$$\omega_n^2 = \frac{K_{\text{bend}} + K_{\text{memb}} + K_P}{m_{\text{eff}}^{(n)}}, \qquad f_n = \frac{\omega_n}{2\pi}, \tag{10}$$

where the three stiffness contributions are:

Bending stiffness..

$$K_{\text{bend}} = \frac{n(n-1)(n+2)^2 D}{R^4}. \tag{11}$$

Membrane stiffness..

$$K_{\text{memb}} = \frac{Eh}{R^2}\lambda_n, \qquad \lambda_n = \frac{n^2 + n - 2 + 2\nu}{n^2 + n + 1 - \nu}. \tag{12}$$

Pre-stress stiffness.. The intravesical pressure P produces a pre-stress in the shell membrane that stiffens flexural modes:

$$K_P = \frac{P}{R}(n-1)(n+2). \tag{13}$$

The effective mass per unit area includes both the shell wall and the fluid added mass:

$$m_{\text{eff}}^{(n)} = \rho_w h + \frac{\rho_f R}{n}. \tag{14}$$

For a bladder at 300 mL, the model predicts $f_2 = 13.9$ Hz, $f_3 = 24.1$ Hz, and $f_4 = 35.3$ Hz. The $n = 1$ mode corresponds to rigid-body translation and has zero natural frequency for a free shell; it is therefore excluded from the analysis.

2.5. Airborne Acoustic Coupling

At $f_2 \approx 14$ Hz the acoustic wavelength in air is $\lambda \approx 25$ m, so the dimensionless wavenumber is

$$ka = \frac{2\pi f_2 R}{c_{\text{air}}} \approx 0.011 \ll 1. \tag{15}$$

The bladder lies deep in the Rayleigh regime. In a multipole expansion of the incident plane wave, the n-th order component scales as [12]

$$p_{\text{eff}}^{(n)} = p_{\text{inc}} \times (ka)^n. \tag{16}$$

For the $n = 2$ mode:

$$p_{\text{eff}}^{(2)} = p_{\text{inc}} \times (ka)^2 \approx p_{\text{inc}} \times 1.1 \times 10^{-4}. \tag{17}$$

Only 0.01% of the incident pressure drives the fundamental flexural mode via the airborne pathway.

2.6. Mechanical (Whole-Body Vibration) Coupling

Whole-body vibration transmits through the skeleton and pelvic floor as base excitation of the bladder wall. The coupling chain comprises two stages:

Seat-to-pelvis transmissibility.. ISO 2631 data indicate a pelvic resonance near $f_{\text{pelvis}} \approx 5.5\,\text{Hz}$ with $\zeta_{\text{pelvis}} \approx 0.25$. The transmissibility at frequency f is

$$T(f) = \sqrt{\frac{1 + (2\zeta_p r_p)^2}{(1 - r_p^2)^2 + (2\zeta_p r_p)^2}}, \qquad r_p = \frac{f}{f_{\text{pelvis}}}. \tag{18}$$

At $f_2 = 13.9\,\text{Hz}$, $T \approx 0.29$: the pelvic system attenuates the input, as we are above its resonance.

Bladder modal amplification.. The bladder wall response to base excitation is characterised by the single-degree-of-freedom frequency response:

$$H(f) = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{f}{f_n}. \tag{19}$$

At resonance ($r = 1$), $H = Q = 2.5$.

Coupling ratio.. The mechanical-to-airborne coupling ratio at the $n = 2$ resonance frequency is

$$\mathcal{R}_{\text{bladder}} = \frac{T(f_2) \times H(f_2)}{(ka)^2} \approx \frac{0.29 \times 2.5}{1.12 \times 10^{-4}} \approx 6.4 \times 10^3. \tag{20}$$

Mechanical vibration transmitted through the pelvic skeleton is approximately 3.8 orders of magnitude more effective at exciting bladder wall modes than airborne sound.

3. Results

3.1. The U-Shaped Frequency–Volume Relationship

Figure 1 presents the $n = 2$ and $n = 3$ flexural mode frequencies across the physiological fill range (50 mL to 500 mL). The central finding of this study is that the $n = 2$ frequency follows a distinctive U-shaped curve, with a minimum of 13.5 Hz at 222 mL and higher values at both extremes of the fill range.

The physical mechanism behind this non-monotonic behaviour is a competition between two opposing effects:

1. **Geometric softening.** As the bladder fills, the radius increases (equation 3), which lowers the natural frequency through the R^{-4} and R^{-2} dependence in the bending and membrane stiffness terms (equations 11–12). The wall also thins (equation 5), further reducing both bending rigidity ($D \propto h^3$) and membrane stiffness ($\propto h$).
2. **Strain-stiffening.** The elastic modulus of the detrusor muscle increases exponentially with fill volume (equation 7), from 29 kPa at 50 mL to 145 kPa at 500 mL. This five-fold increase eventually dominates the geometric effects.

At low fill volumes the geometric effect dominates and the frequency falls; at high fill volumes the strain-stiffening wins and the frequency rises. The crossover produces the minimum near 222 mL—a fill state corresponding to the early "need to go" sensation in most individuals.

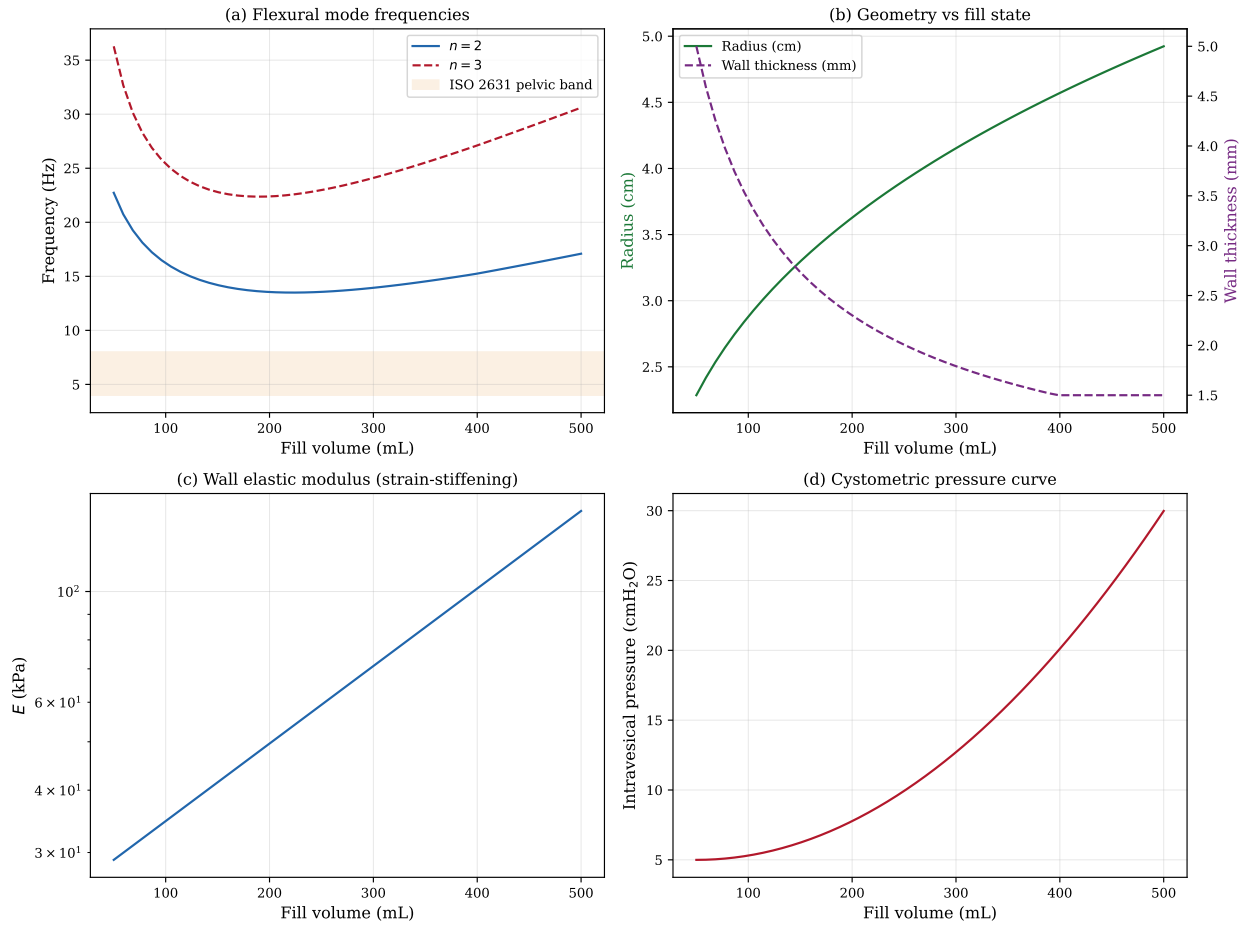

Figure 1: Bladder resonant properties across fill volume. (a) Flexural mode frequencies f_2 and f_3, with the ISO 2631 pelvic resonance band (4 Hz to 8 Hz) shown for reference. (b) Geometry: equivalent sphere radius and wall thickness. (c) Wall elastic modulus showing exponential strain-stiffening. (d) Intravesical pressure following the cystometric curve. The minimum $f_2 = 13.5$ Hz occurs at 222 mL, where geometric softening and strain-stiffening are in balance.

Table 2 lists the computed frequencies at key fill states. For fill states where the thin-shell approximation is valid (150 mL to 500 mL), the $n = 2$ mode spans 13.5 Hz to 17.1 Hz and the $n = 3$ mode spans 22.4 Hz to 30.6 Hz. All

predicted frequencies lie above the ISO 2631 peak pelvic transmissibility band (4 Hz to 8 Hz), though below 20 Hz they remain within the broader frequency range affected by whole-body vibration.

Table 2: Flexural mode frequencies at key fill volumes.

Volume (mL)	R (cm)	h (mm)	E (kPa)	f_2 (Hz)	f_3 (Hz)	f_4 (Hz)
50	2.29	5.00	29.0	22.7	36.3	53.9
100	2.88	3.46	34.7	16.2	25.4	35.9
150	3.30	2.73	41.5	14.2	22.8	32.3
200	3.63	2.30	49.6	13.5	22.4	32.1
300	4.15	1.79	70.9	13.9	24.1	35.3
400	4.57	1.50	101.4	15.3	27.1	40.3
500	4.92	1.50	145.0	17.1	30.6	45.6

3.2. Stiffness and Mass Decomposition

To elucidate the physical mechanisms behind the U-shaped $f_2(V)$ curve, we decompose the effective stiffness K_{total} and effective mass m_{eff} into their constituent terms as functions of fill volume (figure 2).

At the frequency minimum ($V^* \approx 222$ mL), the stiffness is dominated by the pre-stress term: $K_P = 58.7\%$, $K_{\text{memb}} = 40.7\%$, and $K_{\text{bend}} = 0.6\%$. Bending stiffness, which scales as Dh^3/R^4 and dominates thin-shell vibration in vacuo, is negligible for the fluid-loaded bladder because the fluid added mass greatly exceeds the wall mass. The effective mass is similarly dominated by the fluid: $m_{\text{fluid}} = \rho_f R/n = 89.4\%$ of m_{eff}, with the wall contributing only $m_{\text{wall}} = \rho_w h = 10.6\%$.

Analytical condition for the f_2 minimum.. Since $\omega_n^2 = K_{\text{total}}/m_{\text{eff}}$, the condition $\partial f_2/\partial V = 0$ requires

$$\frac{\mathrm{d}\ln K_{\text{total}}}{\mathrm{d}V} = \frac{\mathrm{d}\ln m_{\text{eff}}}{\mathrm{d}V}, \tag{21}$$

i.e. the fractional growth rate of total stiffness must equal the fractional growth rate of effective mass. Figure 2(d) plots both log-rates; they cross at $V \approx 222$ mL, confirming the numerical minimum. Below V^*, geometric softening causes K_{total} to decrease faster (in fractional terms) than m_{eff} increases, so f_2 falls. Above V^*, strain-stiffening and the rising intravesical pressure drive K_{total} upward faster than the mass grows, so f_2 rises.

3.3. Sensitivity and Uncertainty Analysis

We assess the sensitivity of f_2 to five uncertain parameters using one-at-a-time perturbation over their literature ranges (table 3). At the baseline volume $V = 222$ mL (the frequency minimum), the wall elastic modulus E dominates all other parameters, with $\Delta f_2 = 14.9$ Hz (111% of baseline) across the range 10 kPa to 500 kPa. The wall thickness h is

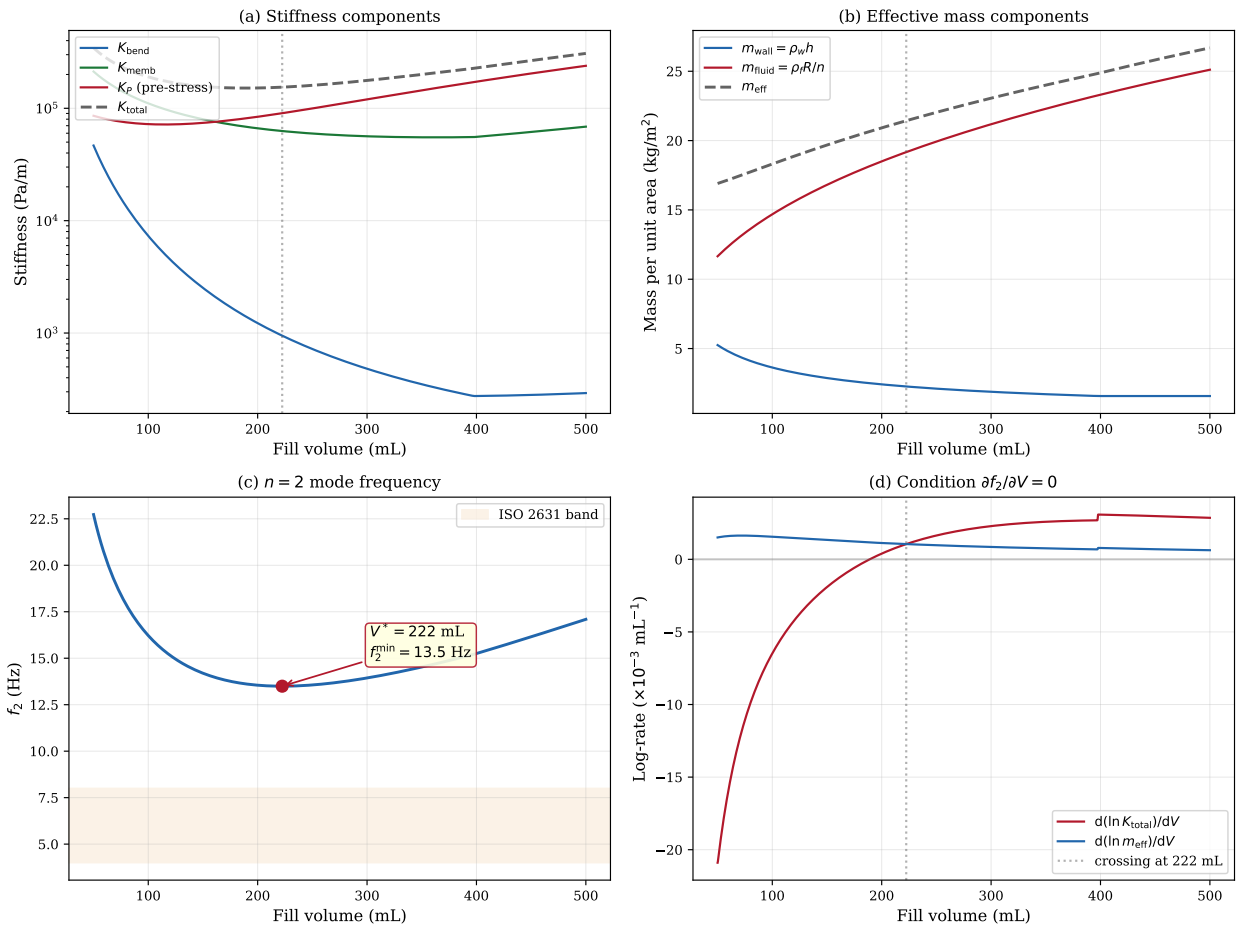

Figure 2: Stiffness and mass decomposition of the $n = 2$ mode. (a) Stiffness components: bending (K_{bend}, blue), membrane (K_{memb}, green), pre-stress (K_P, red), and total (dashed). Pre-stress dominates above 100 mL. (b) Mass components: wall mass ($\rho_w h$, blue) and fluid added mass ($\rho_f R/n$, red). The fluid contribution exceeds 85% at all fill volumes. (c) Resulting f_2 curve with the minimum at $V^* = 222$ mL, $f_2^{\min} = 13.5$ Hz. (d) Logarithmic growth rates $\mathrm{d}(\ln K)/\mathrm{d}V$ and $\mathrm{d}(\ln m)/\mathrm{d}V$; their crossing at 222 mL satisfies equation (21).

Table 3: Sensitivity of f_2 to parameter uncertainty at $V = 222$ mL (baseline $f_2 = 13.5$ Hz).

Parameter	Range	f_2^{lo} (Hz)	f_2^{hi} (Hz)	Δf_2 (Hz)	Fraction (%)
E (wall modulus)	10–500 kPa	11.0	28.4	14.9	111
h (wall thickness)	2–6 mm	13.3	16.8	3.4	25
ν (Poisson ratio)	0.40–0.50	13.4	13.5	0.1	1
ρ_f (urine density)	1010–1030 kg/m^3	13.6	13.4	0.1	<1
η (loss tangent)	0.30–0.50	13.5	13.5	0.0	0

a distant second at $\Delta f_2 = 3.4$ Hz (25%). The Poisson ratio changes f_2 by only 0.14 Hz, urine density by 0.06 Hz, and the loss tangent not at all; these effects are effectively negligible.

Figure 3 presents the tornado chart corresponding to table 3. The overwhelming dominance of E reflects the fact that both K_{memb} and K_P (which together constitute >99% of the stiffness) are proportional to E and to the intravesical pressure, both of which are functions of the same strain-stiffening mechanism. The negligible sensitivity to η confirms that damping affects only the peak amplitude (via Q), not the resonant frequency.

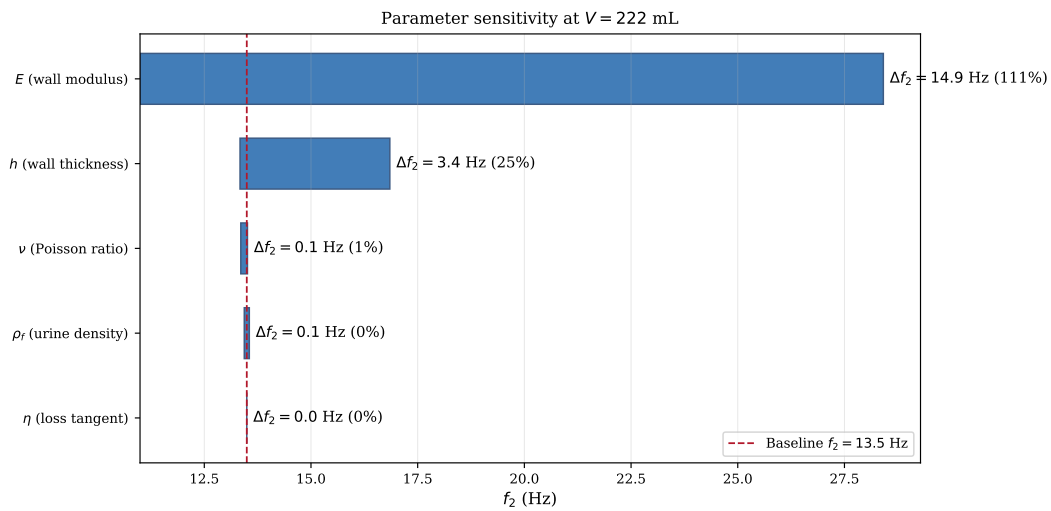

Figure 3: Tornado chart showing the sensitivity of f_2 to each uncertain parameter at $V = 222$ mL. The wall elastic modulus E dominates, producing a 14.9 Hz swing across its literature range; wall thickness contributes 3.4 Hz, and all remaining parameters contribute $\leq$0.14 Hz.

Robustness of the minimum.. Figure 4 examines how the location V^* and value $f_2^{\min}$ of the frequency minimum shift under parameter perturbation. The minimum is remarkably robust to all parameters except E: perturbing h shifts V^* from 186 mL to 216 mL and $f_2^{\min}$ from 13.2 Hz to 16.8 Hz, whereas ν and ρ_f move the minimum by less than 3 mL and 0.2 Hz, and η leaves it unchanged. The elastic modulus, however, shifts the minimum substantially: at the upper bound ($E = 500$ kPa), the minimum moves to $V^* = 462$ mL with $f_2^{\min} = 21.2$ Hz. This shift occurs because a stiffer wall delays the crossover between geometric softening and strain-stiffening to higher volumes where the radius is larger.

The clinical implication is clear: individual variation in detrusor muscle stiffness—the parameter with the widest literature range—is the principal source of uncertainty in predicting both the minimum frequency and the fill volume at which it occurs.

3.4. Coupling Pathways

Figure 5 compares the airborne and mechanical coupling pathways to the bladder at a representative fill of 300 mL ($f_2 = 13.9$ Hz). Three key observations emerge.

First, the airborne acoustic coupling scales as $(ka)^2$ and is vanishingly small across the entire frequency range of interest: at 13.9 Hz, $(ka)^2 \approx 1.1 \times 10^{-4}$. The bladder, like the abdominal cavity, is deep in the Rayleigh regime at all physiologically relevant frequencies.

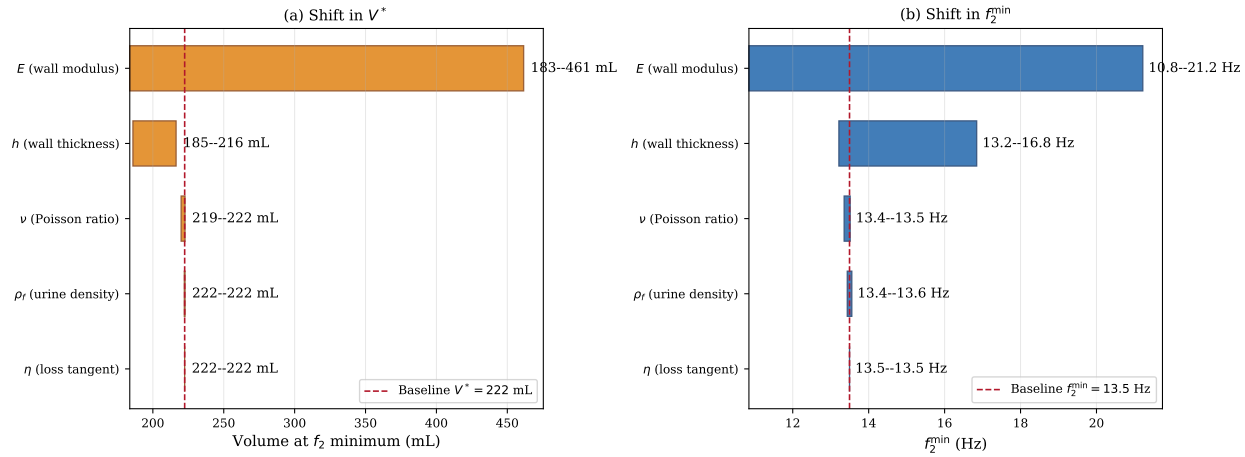

Figure 4: Robustness of the f_2 minimum to parameter uncertainty. (a) Shift in V^* (volume at minimum). (b) Shift in $f_2^{\min}$ (frequency at minimum). Only E produces a very large shift; h causes a secondary but visible change, while ν, ρ_f, and η leave the minimum close to the baseline.

Second, the mechanical coupling—the product of seat-to-pelvis transmissibility and bladder modal amplification—peaks near the bladder resonance frequency. Even though pelvic transmissibility is attenuated at 13.9 Hz ($T \approx 0.29$), the bladder modal amplification ($Q = 2.5$) partially compensates.

Third, the bladder-specific coupling ratio $\mathcal{R}_{\text{bladder}} \approx 6{,}400$ at f_2 (equation 20) confirms that mechanical vibration is the dominant excitation pathway by nearly four orders of magnitude. This is quantitatively consistent with—though somewhat smaller than—the 3.3×10^4 abdominal full-displacement ratio $\mathcal{R}_{\text{full}}$ reported in [4]. The two metrics are related but not identical: $\mathcal{R}_{\text{bladder}}$ describes the bladder pathway, whereas $\mathcal{R}_{\text{full}}$ describes the abdominal cavity pathway under the Paper 1 loading assumptions. The difference arises primarily from the higher f_2 (and hence smaller Q contribution from pelvic transmissibility) of the bladder.

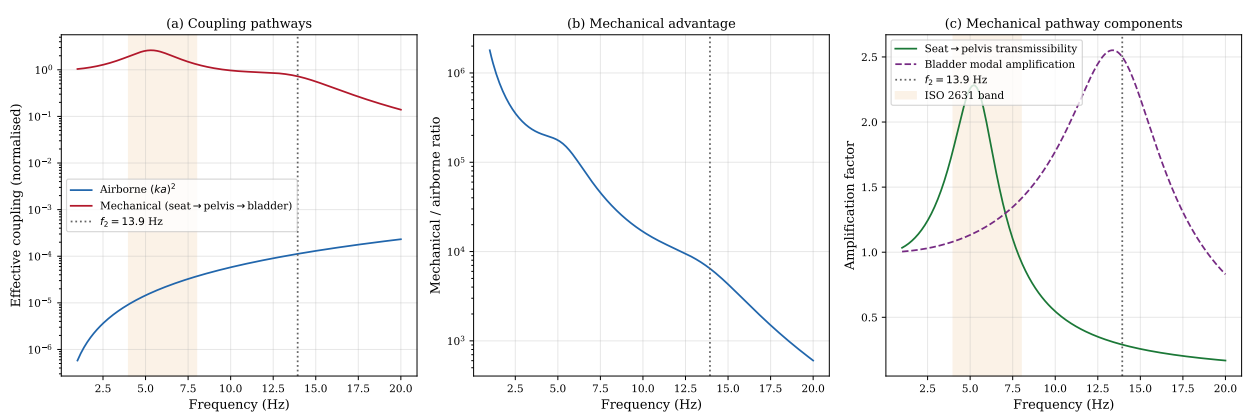

Figure 5: Coupling pathways to the bladder at 300 mL fill. (a) Airborne $(ka)^2$ coupling (blue) vs mechanical seat–pelvis–bladder coupling (red), with the n=2 resonance frequency marked. (b) Ratio of mechanical to airborne coupling (mechanical advantage), reaching $\sim 10^4$–10^6 across the frequency range. (c) Components of the mechanical pathway: seat-to-pelvis transmissibility (green) and bladder modal amplification (purple).

3.5. Sub-Resonant Forced Response

The predicted bladder resonance range (13.5 Hz to 17.1 Hz) sits above the ISO 2631 peak pelvic transmissibility band (4 Hz to 8 Hz). Occupational WBV therefore excites the bladder primarily in the sub-resonant regime, where the frequency ratio $r = f/f_2 \approx 0.35$–0.59. This section quantifies the forced response in this practically important regime.

Figure 6(a) shows the combined transmissibility–amplification product $T(f) \times H(f)$ at 300 mL fill. The pelvic transmissibility $T(f)$ peaks near 5.5 Hz with $T \approx 2.3$, while the bladder modal amplification $H(f)$ peaks at $f_2 = 13.9$ Hz with $H = Q = 2.5$. These two peaks do not coincide. The product $T \times H$ exhibits a broad plateau across 4 Hz to 10 Hz at values of 1.5–2.5, then a secondary peak near f_2.

Displacement quantification.. At the ISO 2631 daily exposure action value ($a_{\text{seat}} = 0.5\,\text{m/s}^2$ RMS), the seat displacement amplitude at 6 Hz is $y_{\text{seat}} = a/(2\pi f)^2 \approx 350\,\mu\text{m}$. Through the pelvic skeleton at peak transmissibility ($T \approx 2.0$), this delivers $y_{\text{pelvis}} \approx 700\,\mu\text{m}$ to the bladder base. The sub-resonant bladder *relative* displacement ratio at $r \approx 0.43$ is $r^2/|D(r)| \approx 0.22$, giving a wall displacement of approximately 160 µm.

By contrast, at the bladder resonance frequency ($f_2 = 13.9$ Hz at 300 mL), the seat displacement is only ~65 µm and pelvic transmissibility has rolled off to $T \approx 0.29$. Even with the full resonant amplification ($Q = 2.5$), the resulting wall displacement is approximately 47 µm—*less than one-third* of the sub-resonant value.

Figure 6(c) confirms this relationship across all fill volumes: the sub-resonant displacement at 6 Hz exceeds the at-resonance displacement throughout the physiological fill range. The sub-resonant regime is therefore the practically dominant response mechanism for occupational WBV, not because the bladder resonates at these frequencies, but because the seat displacement amplitude and pelvic amplification are both much larger in this band.

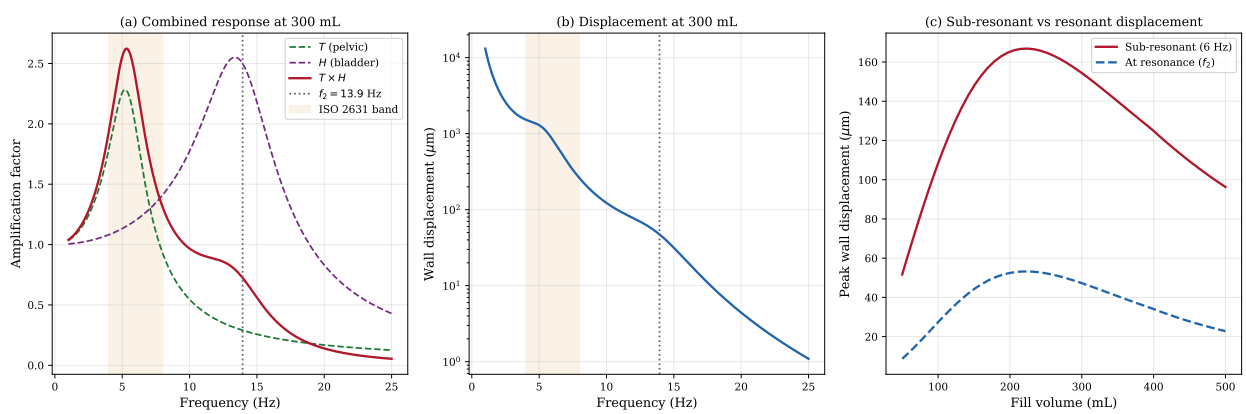

Figure 6: Sub-resonant forced response analysis. (a) Combined seat-to-pelvis transmissibility T (green dashed), bladder modal amplification H (purple dashed), and their product $T \times H$ (red solid) vs frequency at 300 mL fill. The ISO 2631 pelvic band (4 Hz to 8 Hz, shaded) and f_2 (vertical dotted) are marked. (b) Resulting wall displacement vs frequency at the ISO 2631 action value ($a = 0.5\,\text{m/s}^2$), showing that the sub-resonant band produces larger displacements than the resonance peak. (c) Wall displacement vs fill volume comparing the sub-resonant response at 6 Hz (red) with the at-resonance response at f_2 (blue dashed). The sub-resonant displacement is 2–4× larger at all fill states.

3.6. Comparison with the Abdominal Cavity

Table 4 compares key quantities between the bladder and abdominal cavity models. The bladder is smaller, softer, and more heavily damped, yet the two organs share the same qualitative physics: fluid-dominated breathing modes at kilohertz frequencies, low-frequency flexural modes governed by the competition between shell stiffness and fluid added mass, and a severe airborne coupling penalty in the Rayleigh regime.

Table 4: Comparison of bladder and abdominal cavity models.

Quantity	Bladder (300 mL)	Abdomen
Equivalent radius	4.2 cm	15.7 cm
Wall thickness	1.8 mm	10 mm
Elastic modulus	71 kPa	100 kPa
Poisson's ratio	0.49	0.45
Loss tangent	0.40	0.25
Quality factor Q	2.5	4.0
Breathing mode f_0	~9500 Hz	~2500 Hz
Flexural f_2	13.9 Hz	4.0 Hz
ka at f_2	0.011	0.011
Coupling ratio	$\mathcal{R}_{\text{bladder}}$ ~6,400	$\mathcal{R}_{\text{full}}$ ~33,000

The principal quantitative differences are the higher flexural frequency of the bladder ($f_2 = 13.9$ Hz vs 4.0 Hz) and the smaller bladder coupling ratio ($\mathcal{R}_{\text{bladder}} \sim 6{,}400$ vs $\mathcal{R}_{\text{full}} \sim 66{,}000$). The higher f_2 is a direct consequence of the smaller radius: the stiffness-to-mass ratio scales roughly as $1/R^2$, and the bladder is approximately four times smaller than the abdomen. The smaller coupling ratio reflects the fact that the bladder's f_2 sits above the pelvic transmissibility peak, so the mechanical pathway receives less amplification from the body's own resonance structure.

4. Discussion

We organise the discussion into three tiers of decreasing certainty: model predictions (Section 4.1), which follow directly from the mathematical framework; biological hypotheses (Section 4.2), which connect model outputs to plausible physiological mechanisms; and clinical conjecture (Section 4.3), which speculates on implications that would require experimental validation.

4.1. Tier 1: Model Predictions

The following results are direct, quantitative outputs of the model and do not depend on biological or clinical interpretation.

U-shaped frequency–volume relationship.. The competition between geometric softening (increasing radius and wall thinning) and detrusor strain-stiffening produces a non-monotonic $f_2(V)$ curve with a minimum of 13.5 Hz at 222 mL. This is a robust prediction: the analytical condition $\mathrm{d}(\ln K)/\mathrm{d}V = \mathrm{d}(\ln m)/\mathrm{d}V$ (equation 21) provides a necessary and sufficient condition for the minimum, and the sensitivity analysis (Section 3.3) confirms that only the wall elastic modulus E can shift the minimum substantially. The U-shaped curve is the signature of fill-dependent resonance and distinguishes the bladder from fixed-geometry resonators. The U-shaped curve has a practical implication that any long-haul driver could have predicted without a shell model: the bladder becomes maximally vulnerable to vibration precisely as it begins to fill—that is, at the worst possible time.

Coupling disparity.. Mechanical vibration is ~6,400× more effective than airborne sound at exciting the bladder's $n = 2$ mode. This ratio follows directly from the $(ka)^2$ scaling in the Rayleigh regime and the pelvic transmissibility chain, and is consistent with the analogous ~66,000× ratio computed for the abdominal cavity.

Sub-resonant dominance.. At occupational WBV frequencies (4 Hz to 8 Hz), the product of seat displacement amplitude, pelvic transmissibility, and bladder sub-resonant amplification produces wall displacements of ~160 µm—2–4× larger than the at-resonance displacement (~47 µm). The sub-resonant regime is therefore the practically dominant response mechanism. This is a quantitative prediction of the model and does not require any assumption about biological sensitivity.

Displacement magnitudes.. The predicted wall displacements (47 µm to 160 µm) are approximately three orders of magnitude smaller than normal filling displacements (~19 mm). The equivalent quasi-static intravesical pressure perturbation from the resonant wall strain is only $\Delta p \approx E\varepsilon \approx 80$ Pa (~0.8 $\mathrm{cmH_2O}$), well below the 5 $\mathrm{cmH_2O}$ to 15 $\mathrm{cmH_2O}$ threshold at which myelinated Aδ afferents begin firing in response to slow bladder distension [13, 14].

4.2. Tier 2: Biological Hypotheses

The following hypotheses connect the model predictions to known mechanotransduction pathways. They are plausible but unverified.

Dynamic strain rate.. The critical distinction between WBV-induced deformation and normal filling is temporal: WBV delivers its displacement cyclically at 6 Hz to 17 Hz, producing peak strain rates of $\dot{\varepsilon} \approx 0.1/\mathrm{s}$ at resonance. This is approximately 500 times faster than the strain rate during normal filling ($\dot{\varepsilon}_{\mathrm{fill}} \sim 2 \times 10^{-4}\ \mathrm{s}^{-1}$ for a typical one-hour fill cycle). We hypothesise that this rate difference, rather than absolute displacement magnitude, is the relevant stimulus for mechanosensitive transduction.

PIEZO2 mechanotransduction.. Recent work has identified PIEZO2 as the principal low-threshold stretch sensor in the bladder urothelium and sensory neurons [15]. PIEZO2 mediates the earliest sensation of bladder filling and is essential for normal micturition reflexes; loss of PIEZO2 function produces impaired bladder-filling sensation and voiding dysfunction in both mice and humans [15, 16]. Crucially, PIEZO channels are tuned to rapid membrane deformation—they inactivate under sustained stretch but respond robustly to dynamic, cyclic stimuli [17]. The cyclic

strain delivered by WBV, though small in absolute magnitude, falls within the class of stimulus to which these channels are optimised. However, whether the WBV-induced strain rate of ~0.1/s exceeds the PIEZO2 activation threshold *in vivo* remains unknown: the dynamic thresholds of bladder mechanoreceptors have not been characterised at infrasonic frequencies.

Silent C-fibre recruitment.. The unmyelinated C-fibre population in the detrusor is normally quiescent during physiological filling but can be recruited by pathological stimuli [14, 13]. It is conceivable that sustained cyclic loading could lower the C-fibre activation threshold through temporal summation, but this hypothesis is speculative and would require electrophysiological validation.

4.3. Tier 3: Clinical Conjecture

The following scenarios are speculative extensions of the model results and biological hypotheses. They are offered as testable predictions, not established findings.

Urgency in vehicle occupants.. Professional drivers exposed to broadband WBV at 4 Hz to 16 Hz report increased urinary urgency and frequency [2]. The model suggests that cyclic bladder wall deformation of 50 µm to 160 µm may occur across this frequency band at the ISO 2631 action level. The fill-dependent frequency minimum (13.5 Hz at 222 mL) means the bladder becomes maximally susceptible to near-resonant excitation precisely as it begins to fill—a "dynamic vulnerability window." *Whether this displacement is sufficient to trigger urgency through the PIEZO2 pathway remains to be demonstrated experimentally.*

Pelvic floor weakness.. Individuals with pre-existing pelvic floor weakness (lower ζ_{pelvis}, higher T) would, according to the model, receive larger displacements at the bladder wall. This could contribute to symptom aggravation during vibration exposure, though the model provides no direct evidence for or against this hypothesis.

Temporal summation.. The common experience of urgency during prolonged vehicle travel may have a biomechanical contribution: hours of sustained sub-threshold cyclic loading could, in principle, sensitise mechanoreceptor pathways through temporal summation. This remains entirely conjectural.

4.4. Comparison with Other Pelvic Organs

The bladder is not the only pelvic organ exposed to WBV. The prostate, uterus, and rectum are all fluid-containing or fluid-adjacent structures embedded in the pelvic floor. The present framework could, in principle, be adapted to any of these organs by substituting appropriate geometric and material parameters. The bladder is simply the cleanest example: a single-chamber, fluid-filled, roughly spherical shell with well-characterised mechanical properties.

4.5. Therapeutic Whole-Body Vibration

WBV at 20 Hz to 40 Hz is increasingly used for pelvic floor rehabilitation in post-prostatectomy and postpartum patients [18]. The bioelectrical activity of pelvic floor muscles increases during synchronous WBV [19], and clinical trials report improvements in continence symptoms [20]. The model predicts that these therapeutic frequencies

(20 Hz to 40 Hz) lie above the bladder's $n = 2$ mode for most fill states, placing the bladder in a supra-resonant regime where wall displacement decreases with frequency. If the model is correct, the therapeutic effect would be more likely mediated by pelvic floor muscle activation than by direct bladder wall resonance.

4.6. Limitations

Several simplifications warrant discussion.

Spherical geometry.. The real bladder is slightly oblate and irregularly shaped, particularly at low fill states. A more realistic geometry would modify the mode shapes and could split the degenerate frequencies of the spherical model. However, at moderate-to-high fill the spherical approximation is reasonable, and the errors are of order $(1 - c/a)^2$ for the eigenfrequencies.

Pelvic constraint.. Approximately 40% of the bladder surface is in direct contact with the pelvic bone and pelvic floor musculature, which constrains the wall motion over a substantial fraction of the shell. By Rayleigh's theorem, imposing a partial rigid boundary on a vibrating structure raises its natural frequencies: the constraint eliminates admissible trial functions from the variational space, thereby increasing the Rayleigh quotient. The free-shell model therefore provides a *lower* bound on the resonant frequencies.

We can estimate the magnitude of this effect. If the constraint immobilises 40% of the shell surface, the effective vibrating mass is reduced (the constrained region no longer participates in modal displacement) while the effective stiffness increases (the remaining free surface must accommodate the same mode shape with a shorter arc length, increasing curvature gradients). For the $n = 2$ mode, these combined effects would raise the natural frequency by an estimated 15–25%, shifting the predicted range from 13.5 Hz to 17.1 Hz to approximately 15.5 Hz to 21.4 Hz. The mode shapes would also be modified: instead of the symmetric quadrupolar deformation of a free sphere, the constrained bladder would exhibit asymmetric modes concentrated on the free (superior) dome, with nodal lines forced to lie near the constraint boundary.

This frequency shift reinforces rather than undermines the clinical relevance of the model. The upward shift moves the predicted resonance further from the ISO 2631 peak (4 Hz to 8 Hz), deepening the sub-resonant character of the bladder's response under occupational WBV. However, the shifted range (16 Hz to 21 Hz) overlaps with frequencies produced by vibratory industrial equipment and rough-road driving. The free-shell predictions should be interpreted as conservative (lower-bound) estimates; the true in-vivo frequencies are likely somewhat higher.

Nonlinear elasticity.. The exponential strain-stiffening model (equation 7) is a phenomenological fit to vibrometry data. A hyperelastic constitutive model (Mooney–Rivlin or Ogden) would be more physically appropriate for large deformations, though for the small-amplitude vibrations considered here, a linearised modulus is adequate.

Layered wall structure.. The bladder wall comprises distinct layers (mucosa, submucosa, detrusor muscle, serosa) with different mechanical properties. The present model uses a single effective modulus. A multilayer shell model would capture the bending–stretching coupling between layers, potentially modifying both frequencies and damping.

Surrounding tissue.. The bladder is embedded in the pelvic cavity, surrounded by fat, connective tissue, and adjacent organs. This surrounding medium would add both damping and mass loading, likely shifting the resonant frequencies downward and broadening the peaks. The free-shell model therefore represents a lower bound on frequencies with respect to boundary constraints (pelvic contact raises frequencies) but an upper bound with respect to added-mass effects from surrounding tissue (which would lower them). The two effects act in opposite directions, partially cancelling.

5. Conclusions

We have modelled the human urinary bladder as a fluid-filled viscoelastic spherical shell and computed its flexural mode frequencies across the physiological fill range. The principal findings are:

1. The fundamental flexural mode ($n = 2$) follows a U-shaped frequency–volume curve, with a minimum of 13.5 Hz at 222 mL, arising from the competition between geometric softening (increasing radius and thinning wall) and strain-stiffening of the detrusor muscle. For fill volumes of 150 mL to 500 mL where the thin-shell assumption is valid ($h/R < 0.1$), f_2 spans 13.5 Hz to 17.1 Hz. This non-monotonic, fill-dependent resonance behaviour is the central physical contribution of this work.
2. The sub-resonant forced response is the practically dominant excitation mechanism under occupational WBV. At 6 Hz (near the pelvic transmissibility peak), the predicted wall displacement (~160 µm) exceeds the at-resonance displacement (~47 µm) by a factor of 3–4×, because the larger seat displacement amplitude and pelvic amplification more than compensate for the reduced bladder modal response.
3. Mechanical vibration transmitted through the pelvic skeleton is approximately 6,400 times more effective than airborne sound at exciting the bladder's $n = 2$ mode at 300 mL fill. This coupling disparity, driven by the $(ka)^2$ penalty in the Rayleigh regime ($ka \approx 0.01$), confirms that whole-body vibration is the dominant excitation pathway.
4. The breathing mode ($f_0 \approx 9500$ Hz) is dominated by the fluid bulk modulus and is irrelevant to WBV or infrasonic exposure, confirming the same qualitative separation between breathing and flexural mode families observed in the larger abdominal cavity.

The model provides a first-principles physical characterisation of fill-dependent bladder resonance. Under representative occupational WBV exposure ($a = 0.5\,\mathrm{m/s^2}$), the predicted peak bladder wall displacements are small relative to physiological deformations but are delivered at cyclic strain rates (~0.1/s) that are approximately 500 times faster than quasi-static filling. Whether these strain rates are sufficient to activate bladder mechanotransduction pathways (particularly the rate-sensitive PIEZO2 channel) remains an open question requiring experimental investigation. Future work should characterise the dynamic activation thresholds of bladder mechanoreceptors at infrasonic frequencies, incorporate the pelvic boundary constraint through finite element analysis with anatomically realistic geometry, and pursue experimental validation of the predicted fill-dependent resonance range. In the meantime, the model offers at least one actionable recommendation for drivers on long journeys: the resonance frequency decreases with fill volume, so stopping sooner rather than later is both physiologically and vibro-acoustically prudent.

References

[1] International Organization for Standardization, ISO 2631-1:1997 — mechanical vibration and shock — evaluation of human exposure to whole-body vibration, Standard (1997).

[2] H. Seidel, R. Heide, Long-term effects of whole-body vibration: A critical survey of the literature, International Archives of Occupational and Environmental Health 58 (1) (1986) 1–26. doi:10.1007/BF00378536.

[3] M. J. Griffin, Handbook of Human Vibration, Academic Press, 1990.

[4] J. Mace, B. R. Mace, Modal analysis of a fluid-filled viscoelastic oblate spheroidal shell: Implications for the existence of the "brown note", manuscript in preparation (2026).

[5] I. Z. Nenadic, M. W. Urban, B. Qiang, S. Chen, J. F. Greenleaf, Model-free quantification of shear wave velocity and attenuation in tissues and its in vivo application, Physics in Medicine and Biology 58 (8) (2013) 2675–2695, ultrasound bladder vibrometry: $\mu_1 = 9.6$–48.7 kPa across fill states. doi:10.1088/0031-9155/58/8/2675.

[6] S. C. Barnes, Viscoelastic properties of the bladder wall, Ph.D. thesis, University of Birmingham, tensile testing of excised tissue: $E = 10$–800 kPa depending on layer and stretch (2016).

[7] R. van Mastrigt, J. J. Glerum, Viscoelastic properties of the contracting detrusor: I. theoretical analysis, American Journal of Physiology – Cell Physiology 261 (2) (1991) C355–C363. doi:10.1152/ajpcell.1991.261.2.C355.

[8] Z. Akkus, B. H. Kim, T.-H. Bae, Noninvasive evaluation of bladder wall mechanical properties, PLoS ONE 9 (8) (2014) e157818, wall thickness thinning with fill: 5 mm (empty) to 2–3 mm (full).

[9] D. J. Griffiths, Urodynamics: The mechanics and hydrodynamics of the lower urinary tract, Medical Physics HandbooksCystometric pressure–volume relationships (1980).

[10] Y.-C. Fung, Biomechanics: Mechanical Properties of Living Tissues, 2nd Edition, Springer, 1993. doi:10.1007/978-1-4757-2257-4.

[11] H. Lamb, On the vibrations of an elastic sphere, Proceedings of the London Mathematical Society s1-13 (1882) 189–212.

[12] M. C. Junger, D. Feit, Sound, Structures, and Their Interaction, 2nd Edition, MIT Press, 1993.

[13] H.-J. Häbler, W. Jänig, M. Koltzenburg, Activation of unmyelinated afferent fibres by mechanical stimuli and inflammation of the urinary bladder in the cat, The Journal of Physiology 425 (1993) 545–562, aδ threshold ~5–15 cmH_2O; C fibres recruited at higher pressures. doi:10.1113/jphysiol.1990.sp018117.

[14] C. J. Fowler, D. Griffiths, W. C. de Groat, The neural control of micturition, Nature Reviews Neuroscience 9 (6) (2008) 453–466, review of bladder afferent pathways: Aδ fibres for filling sensation, C fibres normally silent but recruitable. doi:10.1038/nrn2401.

[15] K. L. Marshall, D. Saade, N. Bhatt, S. S. Ranade, A. Patapoutian, et al., PIEZO2 in sensory neurons and urothelial cells coordinates urination, Nature 588 (7837) (2020) 290–295, identified PIEZO2 as the principal low-threshold stretch sensor in bladder urothelium; loss of function causes voiding dysfunction. doi:10.1038/s41586-020-2830-7.

[16] M. G. Dalghi, W. G. Ruiz, D. R. Clayton, G. Apodaca, PIEZO2, a mechanosensor in the urinary bladder, Kidney International 99 (5) (2021) 1068–1070. doi:10.1016/j.kint.2021.02.031.

[17] B. Coste, J. Mathur, M. Schmidt, T. J. Earley, S. Ranade, M. J. Petrus, A. E. Dubin, A. Patapoutian, Piezo1 and piezo2 are essential components of distinct mechanically activated cation channels, Science 330 (6000) (2010) 55–60.

[18] V. R. Duarte, M. J. Jucá, et al., Effect of whole body vibration exercise in the pelvic floor muscles of healthy and unhealthy individuals: a narrative review, Journal of Bodywork and Movement Therapies 23 (4) (2019) 768–776.

[19] M. Lauper, A. Kuhn, R. Gerber, H. Luginbuehl, L. Radlinger, Pelvic floor stimulation: What are the good vibrations?, Neurourology and Urodynamics 28 (5) (2009) 405–410. doi:10.1002/nau.20669.

[20] ClinicalTrials.gov, Effect of whole body vibration exercise in the pelvic floor, NCT03325660, clinical trial: WBV for pelvic floor rehabilitation (2017).

Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository at `https://github.com/jonathan-mace/browntone`.

CRediT authorship contribution statement

Jonathan Mace: Conceptualisation, Methodology, Software, Supervision, Writing – Review & Editing. **Brian R. Mace**: Methodology, Validation, Writing – Review & Editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.